

Enterprise Visitor Verification & Access Control Platform

An Summer Internship (2026 – 27)

REPORT FILE

*in partial fulfillment of the requirements for the degree of
Bachelor of Technology*



INDIA GLYCOLS LIMITED

Under the Supervision of:

Mr. Suneet Poothia
(HEAD-IT)
IGL, Kashipur

Submitted by:

Milind Verma
23BCE11695
VIT Bhopal University

Submitted To:

Dr. E. Nirmala
VIT Bhopal University



INDIA GLYCOLS LIMITED

Bonafide Certificate

Certified that this project report titled “ **Enterprise Visitor Verification & Access Control Platform** ” is the bonafide work of **MILIND VERMA - 23BCE11695** who carried out the project work under my supervision.

This project report is submitted for the Project Viva-Voce examination held on 29th June 2026.

Supervisor

Suneet Poothia



INDIA GLYCOLS LIMITED

Declaration of Originality

I hereby declare that this report entitled “Enterprise Visitor Verification & Access Control Platform” represents my original work carried out during my internship at India Glycols Limited (IGL) and submitted as part of the academic requirements of VIT Bhopal University. To the best of my knowledge, this report contains no material previously published or written by another person, nor any material that has been submitted for the award of any other degree, diploma, or qualification at VIT Bhopal University or any other institution.

All sources of information and works of other authors used in this report have been duly acknowledged and cited under the section “References”.

Date: 29th June 2026

Reg. No.: 23BCE11695

Name: MILIND VERMA



INDIA GLYCOLS LIMITED

About India Glycols Limited (IGL)

India Glycols Limited (IGL) is one of India's leading manufacturers of green technology-based chemicals and industrial products. The company specializes in the production of bulk chemicals, specialty chemicals, performance chemicals, natural gums, spirits, industrial gases, sugar, and nutraceutical products.

Established in 1983 with a single Mono-Ethylene Glycol (MEG) manufacturing plant, India Glycols Limited has grown into a diversified and globally recognized organization. Over the years, the company has combined advanced technology, continuous innovation, and a strong commitment to quality to develop a wide portfolio of products that serve various industries across domestic and international markets.

IGL is widely recognized for its focus on sustainability and environmentally responsible manufacturing practices. The company utilizes renewable agricultural resources as raw materials for several of its products, supporting the development of eco-friendly and sustainable solutions. Through continuous research, technological advancement, and customer-focused innovation, India Glycols Limited has established itself as a trusted name in the chemical manufacturing industry.

Today, the company operates multiple manufacturing facilities and serves customers across various sectors, contributing significantly to industrial growth while maintaining its commitment to environmental stewardship and sustainable development.

ACKNOWLEDGEMENT

I would like to express my sincere gratitude to Mr. Sunnet Poothia, my internship supervisor at India Glycols Limited (IGL), for his invaluable guidance, continuous support, and encouragement throughout the development of the project titled “Enterprise Visitor Verification & Access Control Platform.” His technical insights, practical suggestions, and industry expertise greatly contributed to the successful completion of this project. His willingness to share knowledge and provide constructive feedback helped me understand real-world enterprise software development and access management systems.

I would also like to extend my heartfelt thanks to the management and staff of India Glycols Limited (IGL) for providing me with the opportunity to undertake this internship and work on a live industrial project. The professional environment, resources, and cooperation extended by the organization enabled me to gain valuable practical experience and enhance my technical skills.

I am grateful to the faculty members of the School of Computer Science and Engineering (SoCSE), VIT Bhopal University, for their academic guidance and support throughout my academic journey. Their encouragement and continuous motivation helped me successfully apply theoretical concepts to real-world industrial challenges.

I would also like to thank the development communities, researchers, and contributors of open-source technologies such as React.js, FastAPI, TypeScript, Tailwind CSS, SQLAlchemy, and PostgreSQL, whose tools and documentation greatly assisted in the development of this project.

Finally, I express my sincere appreciation to my family, friends, and well-wishers for their constant encouragement, support, and motivation throughout the internship period and the completion of this project.

ABSTRACT

Visitor management and facility access control are critical components of organizational security and operational efficiency. Traditional paper-based visitor registration processes often result in delays, inaccurate records, security vulnerabilities, and difficulties in tracking visitor movements. To address these challenges, this project presents the development of an Enterprise Visitor Verification & Access Control Platform for India Glycols Limited (IGL).

The proposed system is a web-based application designed to streamline and automate the complete visitor management lifecycle, including visitor registration, identity verification, approval workflows, pass generation, check-in/check-out tracking, and reporting. The platform provides a self-service visitor kiosk for quick registration, role-based dashboards for security personnel and administrators, and an approval portal for employees and department heads. The system also supports digital pass generation, photo capture, document management, and real-time visitor monitoring.

Built using React.js, TypeScript, Tailwind CSS, FastAPI, Python, and a relational database, the platform ensures secure data handling through Role-Based Access Control (RBAC) and JWT-based authentication mechanisms. The solution enhances security, reduces manual effort, improves record accuracy, and enables efficient visitor tracking across multiple locations.

The implementation of this system demonstrates how modern web technologies can be leveraged to create a scalable, secure, and user-friendly visitor management solution that supports the operational and security requirements of industrial organizations such as India Glycols Limited.

Index

S.NO	CONTENT	PAGE NO.
1	Introduction	10
1.1	Motivation	11
1.2	Objectives	13
2	Existing Work / Literature Review	15
2.1	Historical Development of Visitor Management Systems	15
2.2	Conceptual Foundations of Visitor Management Systems	16
2.3	Enterprise Web-Based Systems	17
2.4	Authentication and Access Control Technologies	17
2.5	QR Code Technology in Access Management	19
2.6	Database Management in Enterprise Applications	19
2.7	Reporting and Analytics Systems	20
2.8	Applications of Visitor Management Systems	20
2.9	Limitations and Research Gaps	21

2.10	Overall Synthesis	21
3	Topic of the Work	22
3.1	System Design / Architecture	23
3.1.1	Overall Architectural Design Principles	24
3.1.2	Infrastructure Layer	24
3.1.3	Backend Service Architecture	25
3.1.4	Database Architecture	27
3.1.5	Frontend Architecture	28
3.1.6	Component Architecture	29
3.1.7	Security and Access Control Architecture	30
3.1.8	Visitor Lifecycle Workflow	31
3.1.9	Scalability and Deployment Advantages	31
3.2	Working Principle	37
3.3	Results and Discussion	50
3.4	Individual Contribution	54
4	Project Outcome and Applicability	56
4.1	Project Outcome	56
4.2	Applicability of the System	57

5	Conclusion	60
6	References	61
7	Biodata	63

LIST OF FIGURES

FIGURE NO.	FIGURE TITLE	PAGE NO.
Figure 3.1	Infrastructure Architecture Diagram	24
Figure 3.2	Backend API Architecture Diagram	25
Figure 3.3	Database ER Diagram	26
Figure 3.4	Frontend Router Architecture	28
Figure 3.5	Component Architecture Diagram	29
Figure 3.6	RBAC Access Control Diagram	30
Figure 3.7	Visitor Lifecycle Sequence Diagram	31
Figure 3.8	RBAC Access Control Diagram	33
Figure 3.9	Visitor Registration System	39
Figure 3.10	Approval System	40
Figure 3.11	Report Section	47
Figure 3.12	Technical Stack Diagram	49

LIST OF TABLES

TABLE NO.	TABLE TITLE	PAGE NO.
Table 3.1	Authentication and Security Components	33
Table 3.2	Visitor Pass Information	34
Table 3.3	Reporting Features	35
Table 3.4	Deployment Components	36
Table 3.5	Visitor Registration Information	39
Table 3.6	Approval Status Types	41
Table 3.7	Visitor Pass Details	43
Table 3.8	Security Verification Activities	44
Table 3.9	Check-Out Information	46
Table 3.10	Generated Reports	48
Table 3.11	Functional Results Achieved	51
Table 3.12	Performance Observations	51
Table 3.13	Advantages and Future Enhancements	52
Table 4.1	Major Project Outcomes	55
Table 4.2	System Applicability	57

1. Introduction

In modern industrial and corporate environments, ensuring secure and efficient management of visitors is a critical aspect of organizational operations. Large enterprises receive a significant number of visitors, contractors, vendors, candidates, and vehicle operators on a daily basis. Traditional visitor management systems, which rely on manual registers, paper-based approvals, and physical passes, often result in delays, data inaccuracies, security risks, and difficulties in maintaining audit records. As organizations continue to adopt digital transformation initiatives, there is an increasing need for intelligent and automated visitor management solutions.

Security and compliance requirements have become more stringent in industrial organizations. Unauthorized access, incomplete visitor records, and inefficient approval processes can lead to operational disruptions and potential security threats. Manual systems also make it difficult to track visitor movements, generate reports, and maintain historical records for future reference. Therefore, organizations require a centralized platform capable of streamlining visitor registration, verification, approval, and access control processes.

Advancements in web technologies have enabled the development of secure, scalable, and user-friendly enterprise applications. Modern frameworks such as React.js and FastAPI provide the foundation for building high-performance systems that support real-time data processing, role-based access control, and seamless user experiences. By integrating digital workflows with secure authentication mechanisms, organizations can significantly improve operational efficiency while enhancing security.

The Enterprise Visitor Verification & Access Control Platform developed for India Glycols Limited (IGL) is designed to address these challenges. The system provides a comprehensive solution for managing visitors, contractors, vehicles, and employee approvals through a centralized web-based platform. It includes features such as visitor registration, identity verification, digital pass generation, approval workflows, check-in/check-out tracking, role-based dashboards, and reporting capabilities.

The platform follows a multi-user architecture where visitors can register through a self-service kiosk, employees can review and approve visit requests, security personnel can verify passes and monitor entries, and administrators can manage users, departments, and

organizational settings. The system ensures secure data handling through JWT-based authentication and Role-Based Access Control (RBAC), allowing access only to authorized users.

In addition to improving security, the proposed solution enhances operational efficiency by reducing paperwork, minimizing manual intervention, and maintaining accurate digital records. Automated workflows accelerate visitor processing, improve traceability, and provide management with real-time visibility into visitor activities across multiple locations.

In summary, this project leverages modern web development technologies to create a secure, scalable, and efficient visitor management platform. The system serves as a digital transformation initiative for India Glycols Limited, enabling streamlined visitor verification, enhanced access control, and improved organizational security while supporting future scalability and operational excellence.

1.1 Motivation

In today's industrial and corporate environments, ensuring secure and efficient visitor management is essential for maintaining organizational security and operational continuity. Many organizations still rely on traditional paper-based visitor registration processes, manual approval workflows, and physical visitor passes. These conventional methods are often time-consuming, prone to human error, and difficult to monitor, resulting in inefficiencies and potential security risks.

One of the primary motivations behind this project is the need to modernize visitor management processes through digital transformation. Manual systems make it challenging to maintain accurate visitor records, track visitor movements, generate reports, and enforce security policies consistently. Furthermore, paper-based records can be misplaced, damaged, or accessed by unauthorized individuals, compromising data integrity and security.

Another significant motivation is the increasing importance of access control in industrial facilities such as India Glycols Limited (IGL). Organizations regularly receive visitors, contractors, vendors, interview candidates, and vehicle operators. Without a centralized system, managing approvals, verifying identities, and monitoring entries and exits becomes

complex and inefficient. An automated platform can significantly reduce administrative workload while improving transparency and accountability.

The rapid advancement of web technologies and enterprise software solutions has made it possible to develop secure, scalable, and user-friendly applications. Technologies such as React.js, FastAPI, JWT authentication, and relational databases provide powerful tools for building integrated systems capable of handling real-time visitor registration, approval workflows, digital pass generation, and activity tracking.

There is also a growing need for data-driven decision-making in facility management. Organizations require accurate visitor statistics, audit trails, and analytical reports to enhance security measures and optimize operational processes. A centralized visitor management platform can provide valuable insights into visitor trends, access patterns, and compliance requirements.

The motivation of this project is therefore threefold:

- To replace manual visitor registration and approval processes with a secure digital platform.
- To enhance organizational security through visitor verification, access control, and real-time monitoring.
- To improve operational efficiency by automating workflows, reducing paperwork, and maintaining accurate digital records.

By integrating modern web technologies with secure authentication and role-based access control mechanisms, the proposed system aims to provide a reliable, scalable, and efficient visitor management solution that supports the security and operational requirements of India Glycols Limited (IGL).

1.2 Objectives

The primary objective of this project is to design and develop an Enterprise Visitor Verification & Access Control Platform for India Glycols Limited (IGL) that automates visitor registration, verification, approval workflows, and access management while enhancing organizational security and operational efficiency.

The system aims to provide a centralized digital platform for managing visitors, contractors, vendors, candidates, and vehicle entries. By replacing traditional paper-based processes with an integrated web-based solution, the platform seeks to improve accuracy, transparency, and security across the organization.

Specific objectives include:

1. Developing a secure web-based visitor management platform using modern technologies such as React.js, FastAPI, TypeScript, and SQL databases.
2. Creating a self-service visitor registration kiosk that allows visitors to enter their details, capture photographs, and submit visit requests efficiently.
3. Implementing role-based access control (RBAC) to provide secure access for administrators, security personnel, employees, and department heads.
4. Designing an approval workflow system that enables employees and authorized personnel to review, approve, or reject visitor requests digitally.
5. Developing a digital pass generation module with QR code support for visitor identification and verification.
6. Implementing visitor check-in and check-out functionality to maintain accurate records of visitor movements within the facility.
7. Creating centralized dashboards for administrators and security personnel to monitor visitor activities in real time.
8. Providing reporting and analytics features to generate visitor statistics, audit logs, and security reports for management purposes.
9. Ensuring secure authentication and authorization through JWT-based security mechanisms and controlled user permissions.
10. Supporting multi-location and department-wise management to enable scalability across different facilities and operational units.

Another important objective of the project is scalability and maintainability. The system is designed with a modular architecture that allows future enhancements such as biometric verification, mobile application integration, automated notifications, and advanced analytics.

The long-term objective is to establish a secure, efficient, and digitally transformed visitor management ecosystem that strengthens organizational security, improves operational productivity, and supports future technological advancements at India Glycols Limited (IGL).

2. Existing Work / Literature Review

2.1 Historical Development of Visitor Management Systems

Traditional Visitor Management

- Early organizations relied on paper-based visitor registers and manual approval processes.
- Visitors recorded their details in logbooks maintained by security personnel.
- Physical visitor passes and handwritten authorization slips were commonly used.
- These systems provided basic record keeping but lacked security, scalability, and reporting capabilities.

Digital Visitor Management Era

- During the late 1990s and early 2000s, organizations started adopting computerized visitor logging systems.
- Desktop applications replaced paper registers and enabled digital storage of visitor information.
- Barcode-based and electronic identification systems improved tracking accuracy.
- Organizations began integrating visitor data with security and compliance procedures.

Modern Enterprise Visitor Management Platforms

- Cloud-based and web-based visitor management systems emerged with advancements in internet technologies.
- Self-service kiosks, QR code passes, automated approvals, and digital notifications became common features.
- Real-time monitoring, centralized dashboards, and audit reporting improved organizational security.
- Multi-location support enabled enterprises to manage visitor activities across different facilities through a single platform.

Current Industry Trends

- Contactless visitor registration.
- QR code and digital pass verification.
- Mobile-friendly approval workflows.
- Role-Based Access Control (RBAC).
- Integration with enterprise security systems.
- Real-time analytics and reporting.

Visitor management systems have evolved from simple record-keeping tools into intelligent enterprise security and access management platforms.

2.2 Conceptual Foundations of Visitor Management Systems

Visitor Management Systems (VMS) are designed to regulate, monitor, and record visitor activities within an organization while maintaining security and operational efficiency.

Core Components

A modern visitor management system consists of:

- Visitor Registration Module
- Identity Verification Module
- Approval Workflow Engine
- Digital Pass Generation System
- Check-In and Check-Out Tracking
- Security Dashboard
- Reporting and Analytics Module
- Role-Based Access Control (RBAC)

Evolution of Visitor Management

The perception of visitor management systems has evolved over time:

Visitor Register → Digital Logging System → Security Management Platform → Enterprise Access Control Solution

Modern systems focus not only on maintaining records but also on improving security, compliance, and operational efficiency.

2.3 Enterprise Web-Based Systems

Enterprise applications have transformed organizational operations by digitizing workflows and centralizing information management.

Characteristics of Enterprise Systems

- Centralized data storage.
- Multi-user architecture.
- Real-time data access.
- Scalable system design.
- Secure authentication and authorization.

Benefits

- Improved operational efficiency.
- Reduced paperwork and manual effort.
- Better data accuracy and consistency.
- Enhanced reporting and decision-making.

The proposed platform follows enterprise software principles by providing centralized visitor management and access control functionalities.

2.4 Authentication and Access Control Technologies

Authentication and authorization play a crucial role in securing enterprise applications.

Authentication Methods

- Username and Password Authentication
- Multi-Factor Authentication (MFA)
- Token-Based Authentication
- Biometric Authentication

JWT-Based Authentication

JSON Web Tokens (JWT) are widely used for secure user authentication in modern web applications.

Benefits include:

- Stateless authentication.
- Secure session management.
- Scalability.
- Reduced server overhead.

Role-Based Access Control (RBAC)

RBAC restricts system access based on user roles and responsibilities.

Common roles include:

- Administrator
- Security Officer
- Employee
- Department Head
- Visitor

RBAC ensures that users access only the resources relevant to their responsibilities.

2.5 QR Code Technology in Access Management

QR codes have become widely adopted in visitor management and access control systems.

Applications

- Visitor Identification
- Digital Pass Verification
- Contactless Check-In
- Access Validation

Advantages

- Fast scanning process.
- Reduced manual verification.
- Improved accuracy.
- Enhanced visitor experience.

QR code-based passes form an important component of modern visitor management systems.

2.6 Database Management in Enterprise Applications

Database systems provide secure and reliable storage for visitor information and operational records.

Relational Databases

Common enterprise databases include:

- PostgreSQL
- MySQL
- Microsoft SQL Server

Benefits

- Structured data storage.
- Efficient querying.
- Data integrity.
- Scalability.
- Backup and recovery mechanisms.

The proposed system utilizes relational database architecture for secure storage of visitor, employee, approval, and audit data.

2.7 Reporting and Analytics Systems

Reporting modules provide organizations with valuable insights regarding visitor activities and security operations.

Common Reports

- Daily Visitor Reports
- Check-In and Check-Out Logs
- Department-Wise Visitor Statistics
- Audit Trails
- Security Activity Reports

Analytics dashboards support data-driven decision-making and improve organizational transparency.

2.8 Applications of Visitor Management Systems

Visitor management systems are widely used in:

- Industrial Facilities
- Corporate Offices
- Educational Institutions
- Government Organizations
- Research Centers
- Manufacturing Plants
- Healthcare Facilities

These systems help improve security, compliance, and operational efficiency.

2.9 Limitations and Research Gaps

Despite significant advancements, several challenges still exist:

- Dependence on manual verification in many organizations.
- Limited integration between visitor management and access control systems.
- Lack of centralized multi-location visitor management.
- Inadequate real-time monitoring and analytics.
- Security vulnerabilities in poorly designed systems.
- Limited automation of approval workflows.

- Insufficient audit and compliance reporting capabilities.

2.10 Overall Synthesis

- Visitor management systems have evolved from paper-based registers to enterprise security platforms.
- Modern solutions emphasize automation, security, and operational efficiency.
- Web technologies, QR codes, JWT authentication, and RBAC have significantly improved access control systems.
- Reporting and analytics enhance decision-making and compliance management.
- Opportunities remain for developing integrated, scalable, and secure visitor management solutions.

The proposed Enterprise Visitor Verification & Access Control Platform for India Glycols Limited (IGL) addresses these gaps by providing a centralized, secure, and scalable system that automates visitor registration, approval workflows, pass generation, access control, and reporting functionalities.

3. Topic of the Work

The topic of this project is the design and development of an Enterprise Visitor Verification & Access Control Platform for India Glycols Limited (IGL). The project aims to digitize and automate the visitor management process by providing a centralized platform for visitor registration, verification, approval management, access control, and reporting.

The system has been developed as a web-based enterprise application using modern technologies including React.js, TypeScript, Tailwind CSS, FastAPI, Python, and a relational database management system. It is designed to enhance organizational security, improve operational efficiency, and eliminate the limitations associated with traditional paper-based visitor management systems.

The platform provides a self-service visitor registration kiosk where visitors can enter their personal information, purpose of visit, host details, and other required information. The system supports photo capture and document submission to facilitate identity verification and maintain accurate visitor records.

A role-based access control mechanism has been implemented to ensure secure system access. Different user roles such as Visitor, Security Officer, Employee, Department Head, and System Administrator are provided with specific permissions and functionalities according to their responsibilities. JWT-based authentication is used to secure user sessions and protect sensitive information.

The system includes an automated approval workflow that allows employees and authorized personnel to review, approve, or reject visitor requests digitally. Once approved, the platform generates a digital visitor pass containing a QR code that can be used for verification during entry and exit procedures. Security personnel can validate visitor passes and maintain check-in and check-out records through the security dashboard.

A centralized administrative dashboard has been developed to manage users, departments, visitor records, approvals, and organizational settings. The dashboard provides real-time visibility into visitor activities and enables efficient monitoring of access control operations across the organization.

The project also includes reporting and analytics features that generate visitor statistics, access logs, approval histories, and audit reports. These reports assist management in monitoring visitor activities, improving security policies, and maintaining compliance requirements.

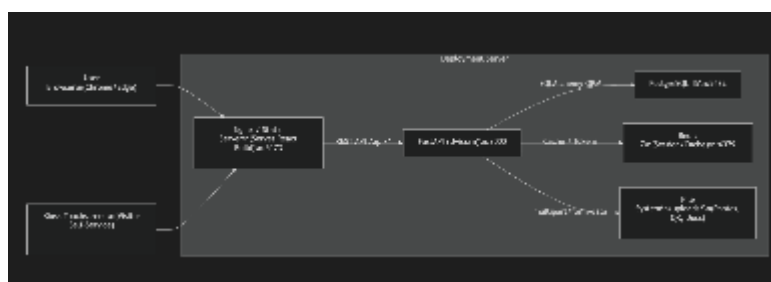
The Enterprise Visitor Verification & Access Control Platform serves as a comprehensive digital solution for India Glycols Limited (IGL). By automating visitor registration, verification, pass generation, approval workflows, and monitoring processes, the system improves security, reduces administrative effort, and provides a scalable framework for future enhancements and organizational growth.

3.1 System Design / Architecture

The Enterprise Visitor Verification & Access Control Platform is designed using a multi-layered enterprise architecture to ensure security, scalability, maintainability, and efficient visitor management operations. The architecture follows a modular approach where each component performs a specific function while remaining integrated with the overall system.

The platform is developed using React.js and TypeScript for the frontend, FastAPI for the backend services, PostgreSQL for database management, Redis for caching and session management, and Docker for containerized deployment. The system supports visitor registration, approval workflows, digital pass generation, QR-code verification, access control, and reporting through a centralized platform.

The architecture is organized into several logical layers including Infrastructure Layer, Backend Services Layer, Database Layer, Frontend Layer, Security Layer, and Visitor Workflow Layer. This separation improves maintainability and allows future expansion without affecting existing functionalities.



[Figure 3.1: Infrastructure Architecture Diagram]

3.1.1 Overall Architectural Design Principles

The architecture is designed based on the following principles:

- Modular system design for easier maintenance and future enhancement.
- Separation of frontend, backend, database, and security responsibilities.
- Role-Based Access Control (RBAC) for secure user management.
- Containerized deployment using Docker.
- Scalable architecture supporting multiple plants and departments.
- Secure communication through REST APIs and JWT authentication.
- Centralized data management and reporting.

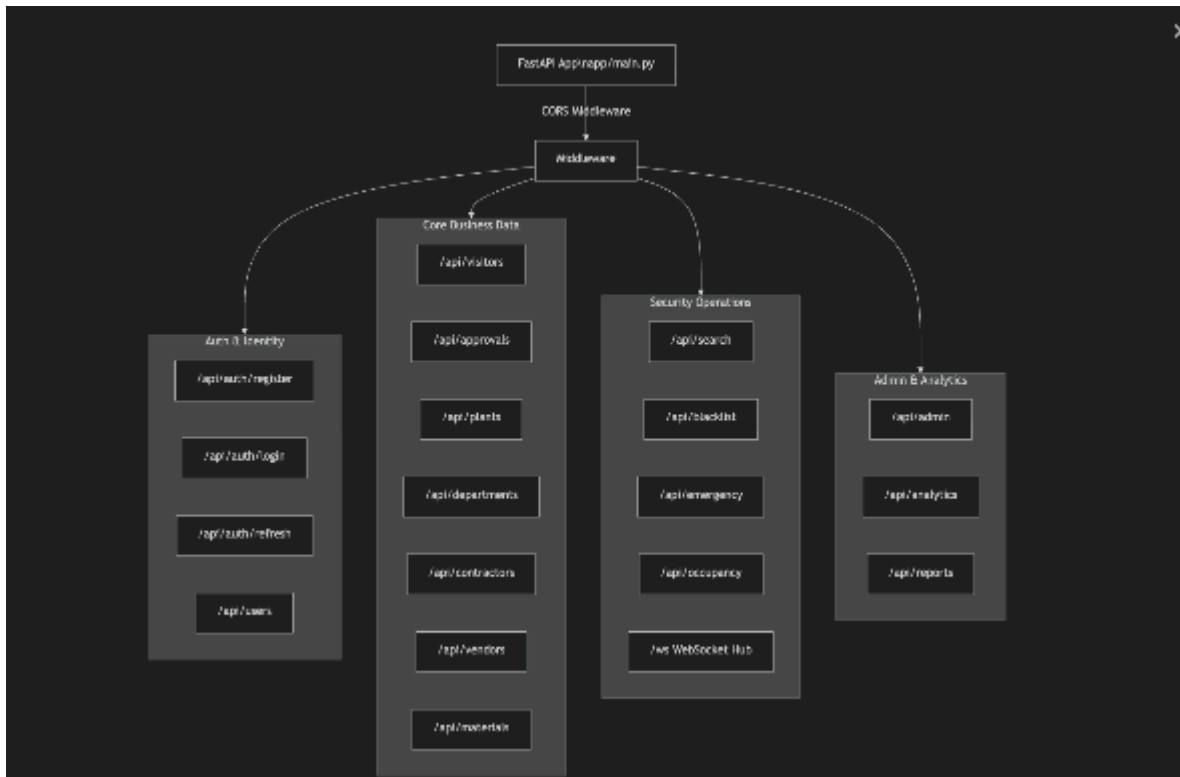
These principles ensure reliability, performance, and long-term scalability of the system.

3.1.2 Infrastructure Layer

The infrastructure layer serves as the foundation of the entire application. Users access the system through a web browser. Incoming requests are routed through Nginx, which acts as a reverse proxy and manages traffic efficiently. Requests are then processed by the FastAPI backend server.

The backend communicates with PostgreSQL for persistent data storage and Redis for caching and performance optimization. Uploaded visitor photographs, documents, and generated passes are stored in the file storage subsystem.

Docker containers are used to package and deploy all services consistently across different environments.



[Figure 3.2: Backend API Architecture Diagram]

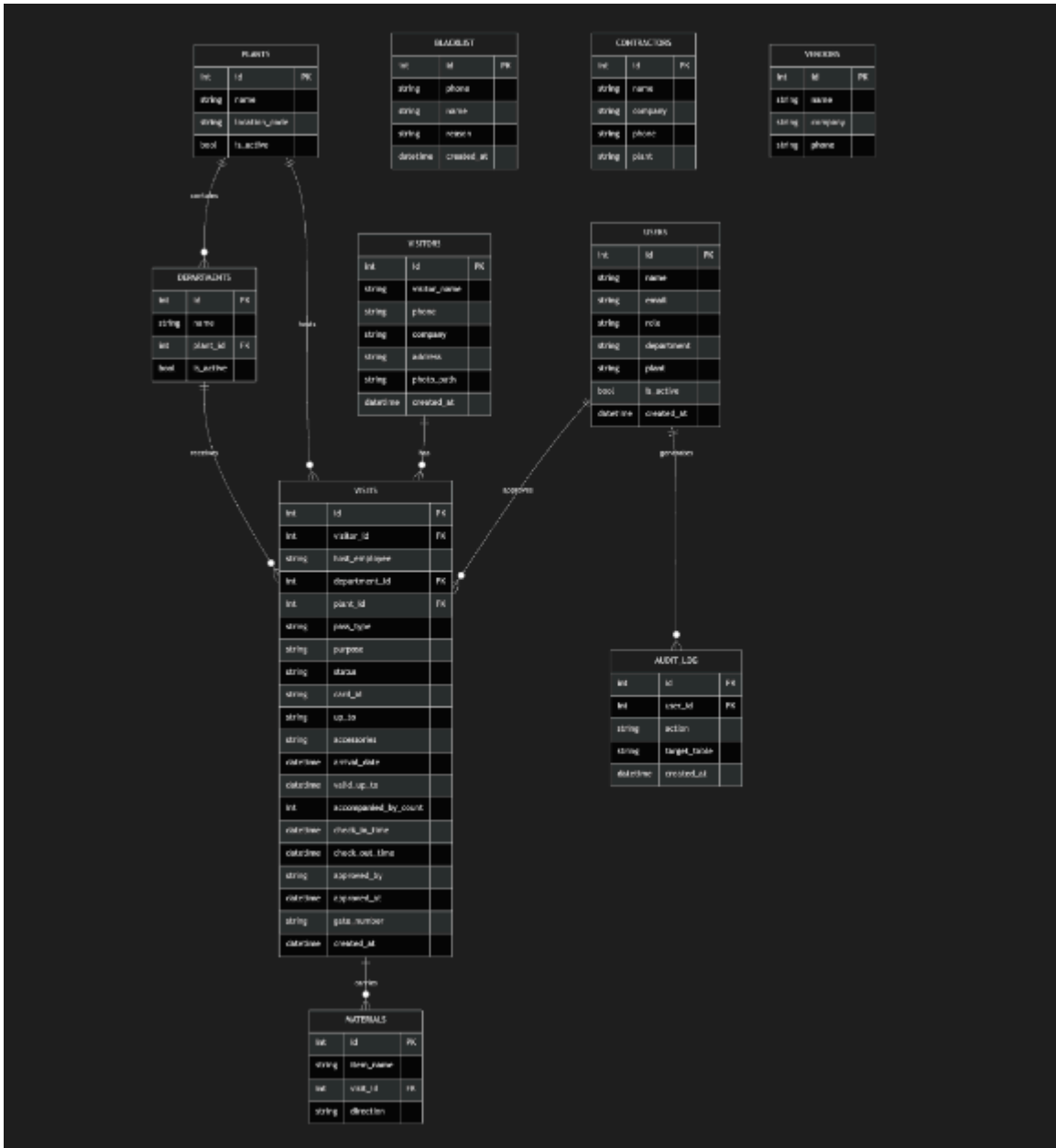
3.1.3 Backend Service Architecture

The backend layer contains the core business logic of the platform. FastAPI provides RESTful APIs for communication between frontend and backend components.

Major service modules include:

- Authentication Service
- Visitor Management Service
- Approval Workflow Service
- Pass Generation Service
- Security Verification Service
- Analytics and Reporting Service
- Plant and Department Management Service
- User and Role Management Service

These services work together to manage the complete visitor lifecycle while ensuring security and data integrity.



[Figure 3.3: Database ER Diagram]

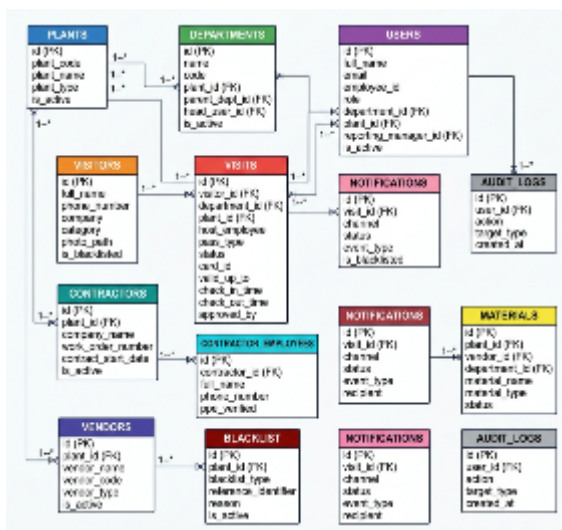
3.1.4 Database Architecture

The database layer stores and manages all operational data within the platform. PostgreSQL is used as the primary relational database.

The database contains information related to:

- Users
- Roles
- Visitors
- Visitor Requests
- Approvals
- Departments
- Plants
- Passes
- Check-In/Check-Out Records
- Audit Logs

Foreign key relationships ensure consistency and maintain data integrity across all modules.



[Figure 3.4: Frontend Router Architecture]

3.1.5 Frontend Architecture

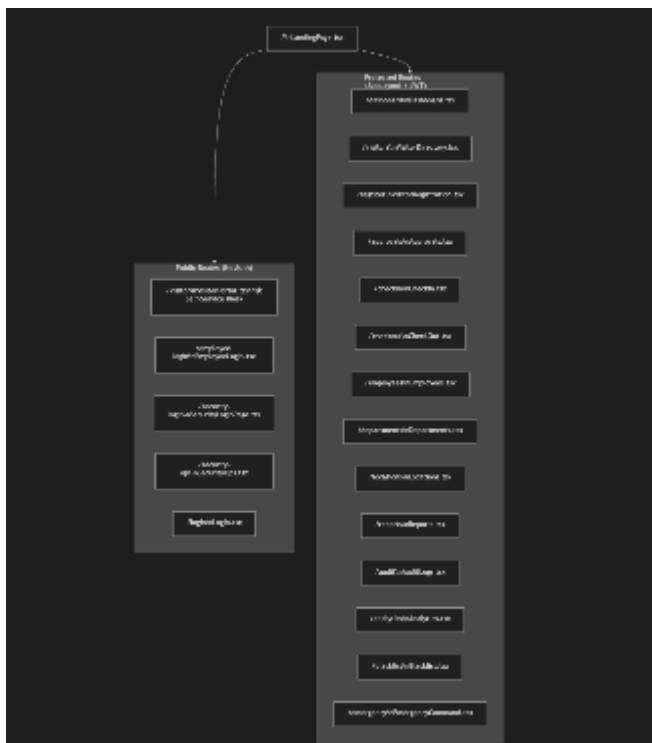
The frontend is developed using React.js, TypeScript, and Tailwind CSS. The user interface is organized using a structured routing system that separates public and protected pages.

The frontend provides:

- Visitor Registration Kiosk

- Login System
- Employee Dashboard
- Security Dashboard
- Administrator Dashboard
- Reports and Analytics Pages
- Plant and Department Management Pages

Protected routes ensure that only authorized users can access sensitive functionalities.



[Figure 3.5: Component Architecture Diagram]

3.1.6 Component Architecture

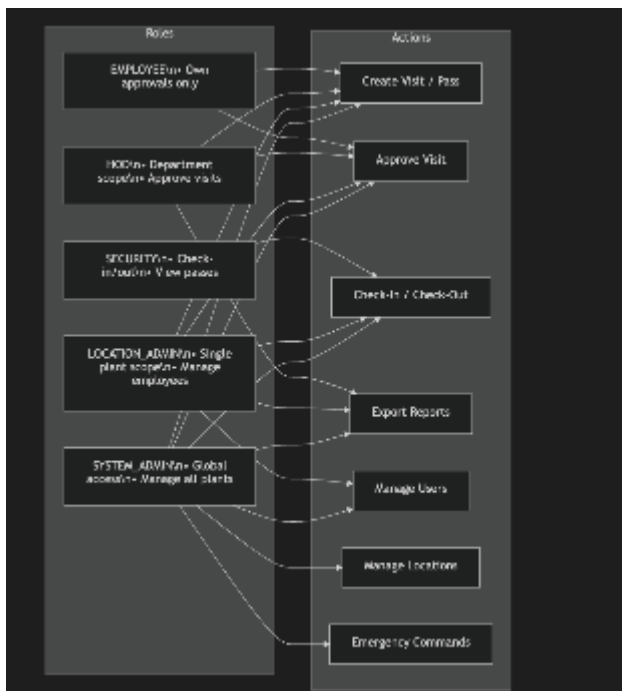
The platform consists of reusable frontend components that improve maintainability and user experience.

Major components include:

- Enterprise Pass Generation Engine
- QR Code Generator

- Visitor Registration Forms
- Dashboard Widgets
- Theme Management System
- Print Management Module
- Notification Components
- Reusable UI Elements

This component-based architecture enables rapid development and easier maintenance.



[Figure 3.6: RBAC Access Control Diagram]

3.1.7 Security and Access Control Architecture

Security is implemented using JWT-based authentication and Role-Based Access Control (RBAC).

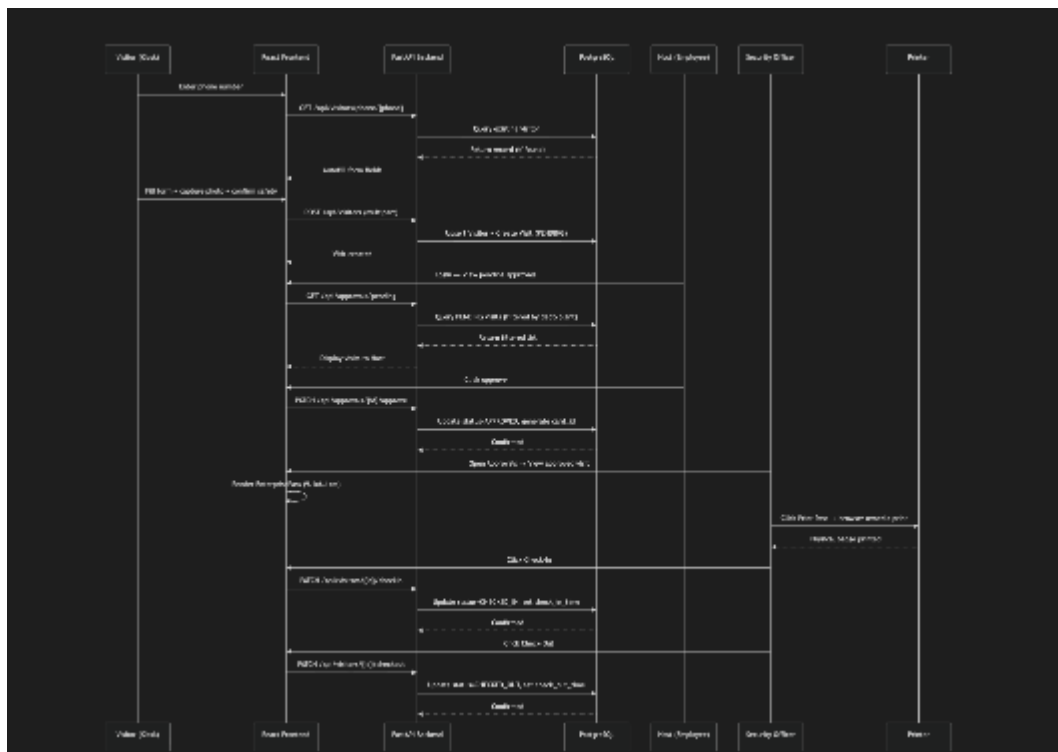
The system defines five major user roles:

- System Administrator
- Plant Administrator
- Department Head

- Employee
- Security Officer

Each role is assigned specific permissions to ensure controlled access to resources and functionality.

Additional security measures include:



[Figure 3.7: Visitor Lifecycle Sequence Diagram]

3.1.8 Visitor Lifecycle Workflow

The visitor lifecycle workflow represents the complete operational flow of the platform.

The process begins when a visitor registers through the kiosk interface and submits visit information. The request is forwarded to the concerned employee or department for approval. Once approved, the system generates a QR-code-enabled visitor pass.

Security personnel verify the pass during entry and record visitor check-in information. Upon completion of the visit, the visitor checks out through the security desk, and the system records the exit details. All activities are logged and available for reporting and auditing purposes.

3.1.9 Scalability and Deployment Advantages

The modular architecture provides several advantages:

- Independent module maintenance.
- Easy addition of new features.
- Multi-location deployment support.
- Docker-based deployment consistency.
- Future cloud deployment capability.
- Integration with biometric devices and access control hardware.
- Support for enterprise-scale visitor management operations.

These features ensure that the platform remains scalable, reliable, and adaptable to future organizational requirements.

3.1.10 Authentication and Security Architecture

Security is one of the most critical aspects of the Enterprise Visitor Verification & Access Control Platform. The system implements a secure authentication and authorization mechanism to ensure that only authorized users can access system resources. Authentication is implemented using JSON Web Tokens (JWT), which provides a secure and stateless method for user session management.

When a user logs into the system, the backend validates the credentials and generates a JWT token. This token is attached to subsequent requests and verified before granting access to protected resources. The token-based authentication mechanism improves security while reducing server-side session management overhead.

The system also implements Role-Based Access Control (RBAC), where permissions are assigned according to user roles. Different users such as System Administrators, Plant Administrators, Department Heads, Employees, and Security Officers are granted access only to the functionalities required for their responsibilities.

Additional security features include encrypted password storage, secure HTTPS communication, API validation, audit logging, and access monitoring. These measures ensure confidentiality, integrity, and availability of organizational data.

Role	Hierarchy Level	Register Visit	Approve Visit	Check-In/Out	Manage Employees	Manage Locations	View Reports	Emerg Contr
CORPORATE_SUPER_ADMIN	1 (Highest)	✓	✓	✓	✓	✓	✓ (All Plants)	✓
PLANT_ADMIN	2	✓	✓	✓	✓	✓ (Own Plant)	✓ (Own Plant)	✓
DEPARTMENT_HEAD	3	✓	✓ (Own Dept)	✗	✗	✗	✓ (Own Dept)	✗
HR_MANAGER	4	✓	✓	✗	✓	✗	✓	✗
HR_EXECUTIVE	4	✓	✗	✗	✓	✗	✓	✗
RECEPTIONIST	5	✓	✗	✗	✗	✗	✗	✗
SECURITY_SUPERVISOR	6	✗	✗	✓	✗	✗	✓	✓
SECURITY_GUARD	7	✗	✗	✓	✗	✗	✗	✗
EMPLOYEE	8 (Lowest)	✓	✓ (Own Visits)	✗	✗	✗	✗	✗

[FIGURE 3.8 – RBAC ACCESS CONTROL DIAGRAM]

Table 3.1: Authentication and Security Components

Component	Purpose
JWT Authentication	Secure user authentication
RBAC	Role-based authorization
HTTPS	Secure communication
Password Encryption	Protect user credentials
Audit Logs	Activity monitoring
Session Validation	Prevent unauthorized access

3.1.11 Visitor Pass Generation System

The Visitor Pass Generation System is responsible for generating digital visitor passes after approval of visitor requests. The generated pass contains essential information including visitor details, host information, visit date, department, and unique QR code.

The QR code serves as a digital identifier that enables quick verification by security personnel. The pass generation module automatically creates print-ready visitor passes that can be downloaded or printed directly from the system.

The system supports customized pass templates and maintains records of all generated passes for future reference and auditing purposes.

Table 3.2: Visitor Pass Information

Field	Description
Pass ID	Unique visitors pass number
Visitor Name	Name of visitor
Host Name	Employee being visited
Department	Department details
Visit Date	Scheduled visit date
QR Code	Unique verification code
Status	Approved / Rejected / Expired

3.1.12 Reporting and Analytics Architecture

The reporting module provides management with detailed insights into visitor activities, approvals, and access control operations. The system generates both real-time and historical reports to support security monitoring and decision-making.

The analytics dashboard displays visitor trends, department-wise visitor statistics, approval rates, security activity logs, and plant-wise visitor distribution. Reports can be filtered based on date ranges, departments, visitor types, and approval status.

Generated reports can be exported in PDF and spreadsheet formats for documentation and compliance purposes.

Table 3.3: Reporting Features

Report Type	Description
Visitor Report	Visitor statistics
Approval Report	Approval activities
Security Report	Check-in/check-out logs
Department Report	Department-wise visits
Audit Report	System activities

3.1.13 Docker Deployment Architecture

To ensure consistency across different environments, the platform is deployed using Docker containerization technology. Docker packages the frontend, backend, database services, and supporting components into isolated containers.

Containerization simplifies deployment, reduces dependency conflicts, and enables rapid scaling of system services. Docker Compose is used to manage multiple containers and establish communication between services.

The deployment architecture supports local development, testing environments, and future cloud deployment scenarios.

Table 3.4: Deployment Components

Component	Technology
Frontend Container	React + Nginx
Backend Container	FastAPI
Database Container	PostgreSQL
Cache Container	Redis
Container Manager	Docker Compose
Reverse Proxy	Nginx

3.1.14 System Advantages

The Enterprise Visitor Verification & Access Control Platform provides several organizational benefits.

- Eliminates paper-based visitor registration.
- Improves security through QR-based verification.
- Supports role-based access control.
- Enables centralized visitor monitoring.
- Generates real-time reports and analytics.
- Supports multi-plant deployment.
- Simplifies visitor approval workflows.
- Enhances audit and compliance management.

The modular architecture also ensures future scalability and integration with additional enterprise systems.

3.2 Working Principle

The working principle of the Enterprise Visitor Verification & Access Control Platform is based on a centralized digital workflow that automates visitor registration, approval, verification, access control, and reporting. The system replaces traditional paper-based visitor management processes with a secure and efficient web-based platform.

The workflow begins when a visitor enters their information through the Visitor Registration Kiosk. The submitted information is processed by the backend system and forwarded to the concerned employee or department for approval. Once approved, a QR-code-enabled visitor pass is generated. Security personnel verify the pass during entry and exit procedures, while all activities are logged for auditing and reporting purposes.

The complete workflow ensures improved security, real-time monitoring, and efficient visitor management across the organization.

3.2.1 Visitor Registration

The visitor registration process serves as the entry point of the system. Visitors enter their personal information, contact details, organization name, purpose of visit, host employee information, and other required details through the kiosk interface.

The system validates the entered information before storing it in the database. Visitor photographs and supporting documents can also be captured and uploaded during registration.

Key Activities

- Visitor enters personal information.
- Purpose of visit is specified.
- Host employee details are selected.
- Visitor photograph is captured.
- Request is submitted for approval.

The screenshot shows a web-based form for visitor registration at Indian Glycol Limited. The form is titled "Security Clearance Request" and includes a "Please verify your details before submission" note. It has a "NEXT/BACK TOOL" button. The form is divided into several sections: "Personal Details" (Title, Full Name, Contact Number, Email), "Visit Details" (Location (Plant), Department, Host Employee, Purpose), "Clearance Details" (Clearance Up To, Category, Arrival (DT), Valid Up To (DT)), "Accompanied By" (ID Co-Sight, Accompanied By, Token ID (Optional)), "Full Address", and "Declared Assets (Laptops, Tools)". There is also an "Identity Scan" section with a camera view and a "START CAM" button. At the bottom, there is a "SAFETY DECLARATION" section with a "I AGREE & ACCEPT" checkbox and a "GENERATE PASS" button.

[FIGURE 3.9 – Visitor Registration System]

Table 3.5: Visitor Registration Information

Field	Description
Visitor Name	Name of visitor
Mobile Number	Contact information
Organization	Company or organization
Purpose	Reason for visit
Host Employee	Employee being visited
Visit Date	Scheduled date
Photograph	Visitor image

3.2.2 Approval Workflow

After registration, the visitor request is forwarded to the concerned employee or department head for approval.

The approver can review the visitor's details and either approve or reject the request. Notifications are generated automatically to inform relevant users about approval status.

Approval Actions

- Review visitor requests.
- Verify visit purpose.
- Approve or reject requests.
- Generate approval records.

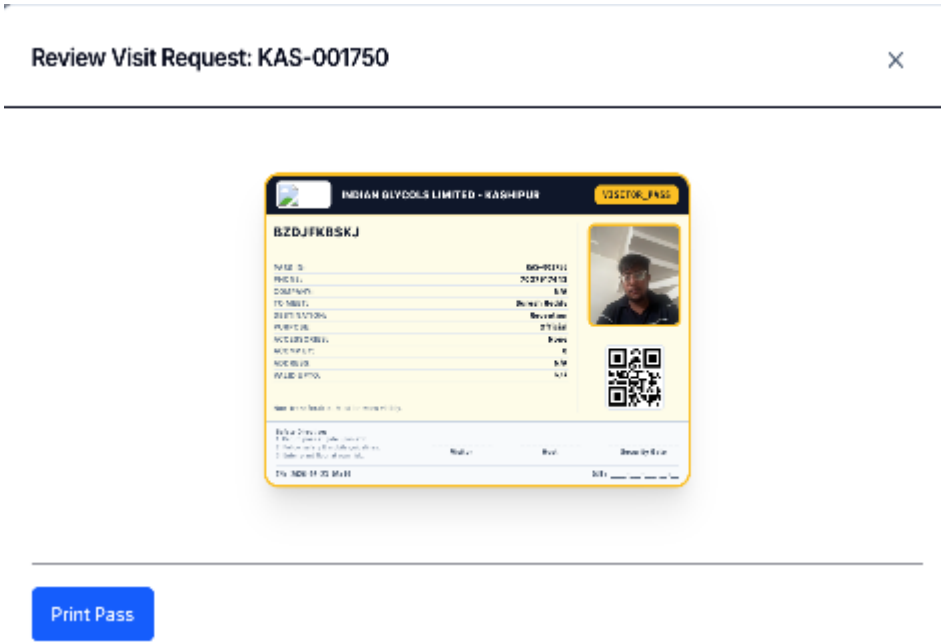
VISITOR ID	VISITOR NAME	COMPANY	PURPOSE	DEPARTMENT	HOST EMPLOYEE	STATUS	ACTIONS
KAS-001752	Irod[Rishik]	N/A	Official	Reception	Suresh Reddy [HR MANAGER, Reception]	APPROVED	Review
KAS-001748	Milind Verma	N/A	Official	Finance	vikram.kumar958@indianoil.in	APPROVED	Review
KAS-001748	Milind Verma	N/A	Official	Sales	deepa.lyer100@indianoil.in	APPROVED	Review
KAS-001747	Milind Verma 28BCE1666	N/A	Personal	Production	Rishik Verma [EMPLOYEE, Production]	PENDING	Review
DEI-001345	Sneha Iyer	Consult Solutions Pvt Ltd	Audit	Maintenance	Kavita Reddy (kavita.reddy244@indianoil.in)	PENDING	Review
KAS-001343	Manoj Verma	Bulk Solutions Pvt Ltd	Official	Finance	Amit Jain (amit.jain151_000@indianoil.in)	EXPIRED	Review
MOI-001344	Arun Singh	Consult Solutions Pvt Ltd	Vendor	Therapist	Anjali Gupta (anjali.gupta151@indianoil.in)	EXPIRED	Review
MOI-001343	Kavita Singh	Tech Solutions Pvt Ltd	Meeting	Finance	Advik Dubey (advik.dubey151@igil.com)	PENDING	Review

[FIGURE 3.10 – Approval System]

The QR code serves as a secure identifier that can be verified by security personnel during entry and exit operations.

Pass Contents

- Pass ID
- Visitor Information
- Host Information
- Visit Date
- Department
- QR Code



[FIGURE 3.9 – Visitor Approved Pass]

Table 3.7: Visitor Pass Details

Component	Purpose
Pass ID	Unique identifier
QR Code	Verification mechanism

Visitor Name	Identification
Department	Destination area
Validity Period	Access duration

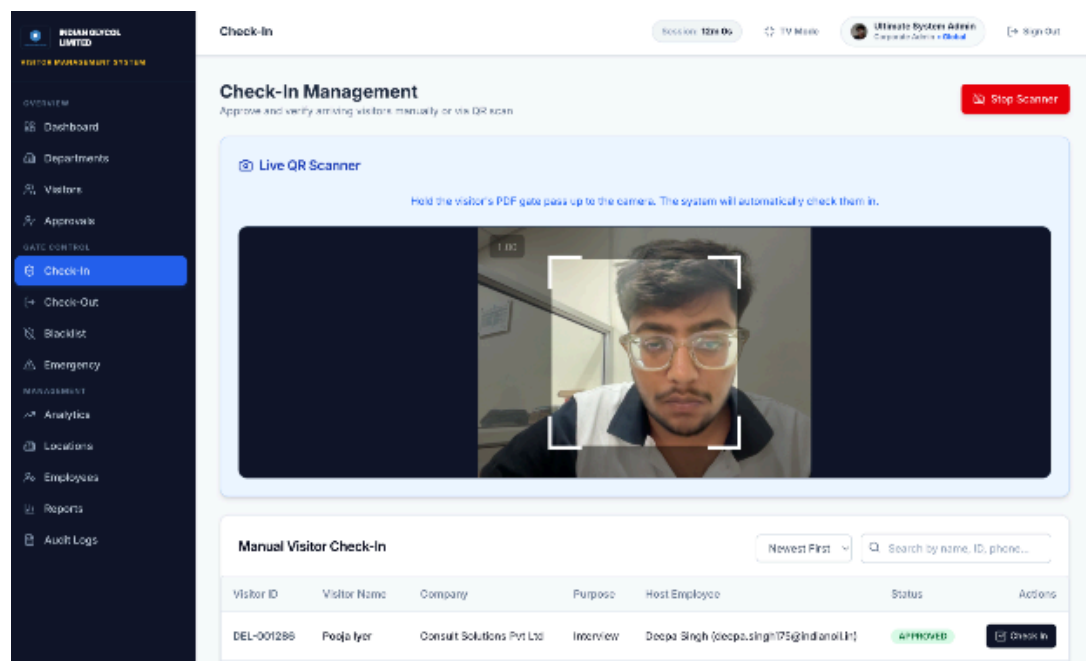
3.2.4 Security Verification and Check-In

When the visitor arrives at the facility, security personnel verify the QR code using the security dashboard.

The system validates the pass and records visitor entry information. Unauthorized or expired passes are automatically rejected.

Security Operations

- QR code scanning.
- Visitor identity verification.
- Entry authorization.
- Check-in recording.



[FIGURE 3.9 – Visitor QR CheckIn System]

Table 3.8: Security Verification Activities

Activity	Description
QR Scan	Pass verification
Identity Check	Visitor validation
Check-In	Entry recording
Monitoring	Real-time tracking

3.2.5 Visit Monitoring

During the visit, the system maintains records of active visitors within the facility.

Administrators and security personnel can view real-time visitor information through the dashboard.

Monitoring Features

- Active visitor list.
- Department-wise visitor tracking.
- Visitor history.
- Real-time status updates.

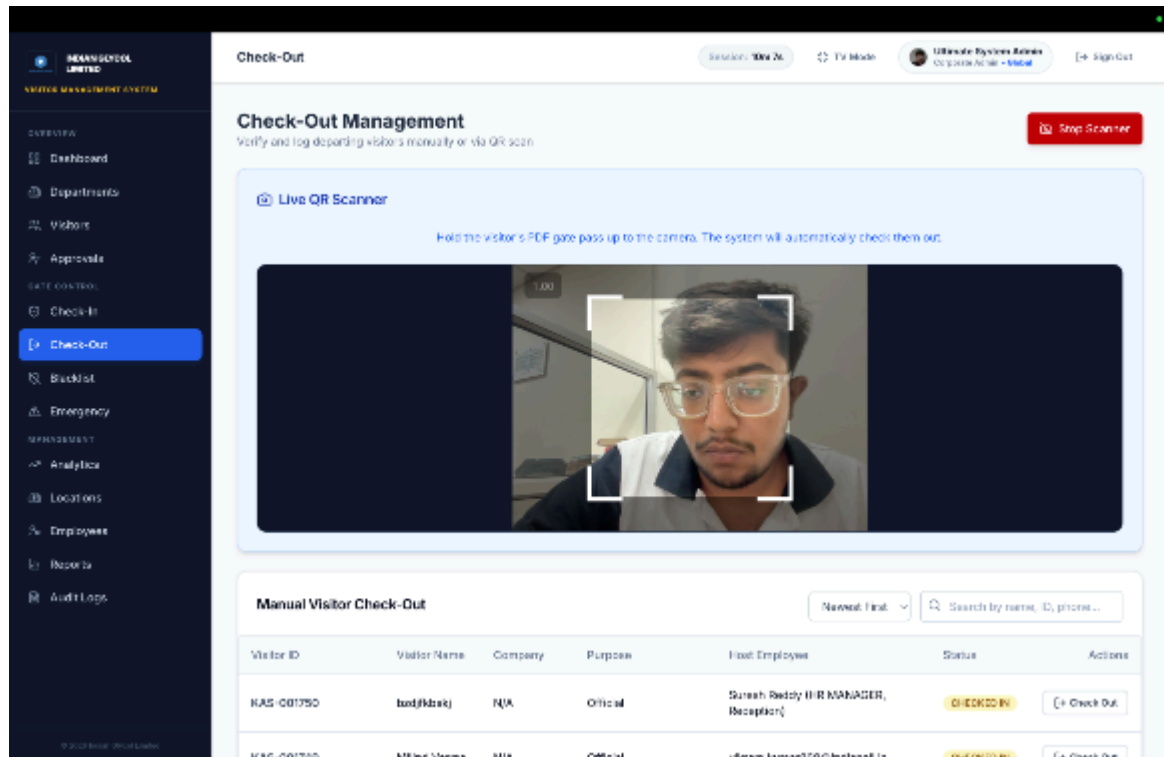
3.2.6 Check-Out Process

At the end of the visit, security personnel record the visitor's exit through the check-out module.

The system updates visitor status and stores departure information for future reference.

Check-Out Activities

- Exit verification.
- Check-out recording.
- Pass closure.
- Audit log generation.



[FIGURE 3.9 – Visitor QR CheckOut System]

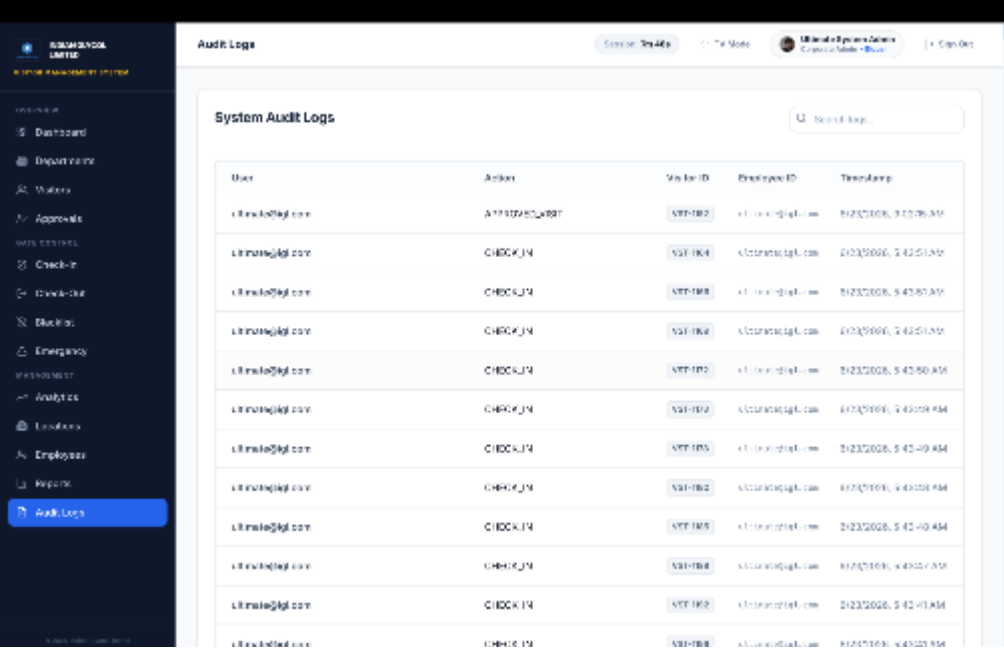
Table 3.9: Check-Out Information

Field	Description
Visitor ID	Visitor reference
Exit Time	Departure timestamp
Security Officer	Verification personnel
Status	Completed visit

3.2.7 Reporting and Audit Logging

All visitor activities are automatically recorded within the system.

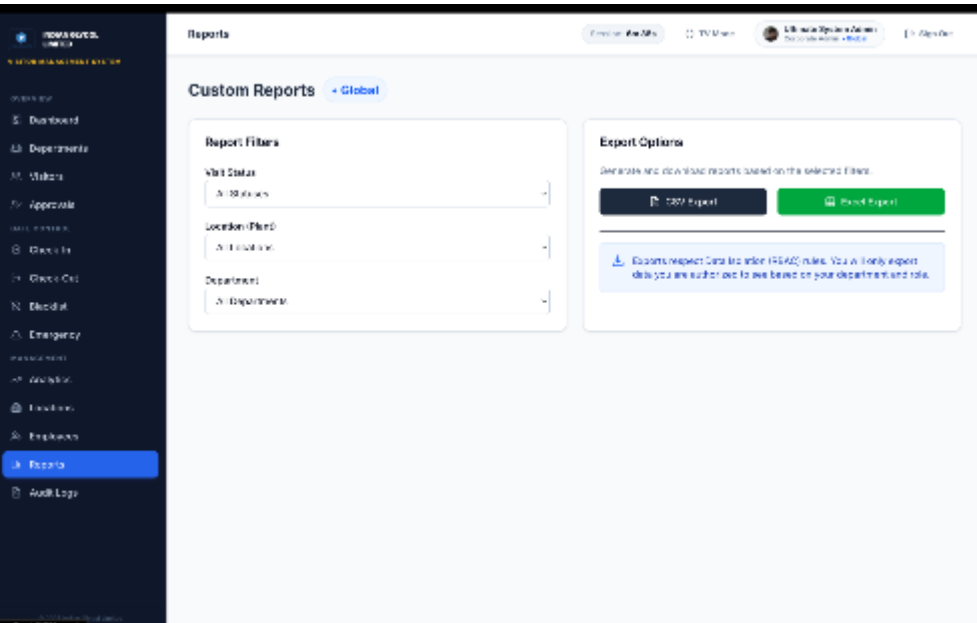
The reporting module generates visitor statistics, approval histories, check-in/check-out logs, and audit reports. These reports help management monitor visitor activity and maintain security compliance.



The screenshot displays the 'Audit Logs' section of a software interface. It features a sidebar on the left with navigation options like Dashboard, Department, Visitors, Approvals, Check-in, Check-out, Blocklist, Emergency, Analytics, Locations, Employees, and Reports. The 'Audit Logs' option is highlighted. The main content area is titled 'System Audit Logs' and contains a table with the following columns: User, Action, Visitor ID, Employee ID, and Timestamp. The table lists several log entries for the user 'ulimale@lgl.com' performing 'CHECK_IN' actions at various timestamps in 2026.

User	Action	Visitor ID	Employee ID	Timestamp
ulimale@lgl.com	ATTENDANCE LOG	NTF-182	ulimale@lgl.com	2/23/2026, 5:43:06 AM
ulimale@lgl.com	CHECK_IN	NTF-184	ulimale@lgl.com	2/23/2026, 5:43:51 AM
ulimale@lgl.com	CHECK_IN	NTF-185	ulimale@lgl.com	2/23/2026, 5:43:57 AM
ulimale@lgl.com	CHECK_IN	NTF-186	ulimale@lgl.com	2/23/2026, 5:43:58 AM
ulimale@lgl.com	CHECK_IN	NTF-182	ulimale@lgl.com	2/23/2026, 5:43:58 AM
ulimale@lgl.com	CHECK_IN	NTF-182	ulimale@lgl.com	2/23/2026, 5:43:58 AM
ulimale@lgl.com	CHECK_IN	NTF-185	ulimale@lgl.com	2/23/2026, 5:43:58 AM
ulimale@lgl.com	CHECK_IN	NTF-182	ulimale@lgl.com	2/23/2026, 5:43:58 AM
ulimale@lgl.com	CHECK_IN	NTF-185	ulimale@lgl.com	2/23/2026, 5:43:58 AM
ulimale@lgl.com	CHECK_IN	NTF-182	ulimale@lgl.com	2/23/2026, 5:43:58 AM
ulimale@lgl.com	CHECK_IN	NTF-182	ulimale@lgl.com	2/23/2026, 5:43:58 AM
ulimale@lgl.com	CHECK_IN	NTF-182	ulimale@lgl.com	2/23/2026, 5:43:58 AM

[FIGURE 3.10 – Audit Logs]



The screenshot shows the 'Reports' section of the software interface. It includes a sidebar with navigation options similar to the previous figure. The main content area is titled 'Custom Reports' and features a 'Global' filter. Below this, there are two main sections: 'Report Filters' and 'Export Options'. The 'Report Filters' section contains three dropdown menus for 'Visit Status' (set to 'All Statuses'), 'Location Filter' (set to 'All Locations'), and 'Department' (set to 'All Department'). The 'Export Options' section includes two buttons: 'CSV Export' and 'Excel Export'. Below these buttons, there is a note stating: 'Reports respect data log order (15000) rules. You can only export data you are authorized to see based on your department and role.'

[FIGURE 3.11 – Report Section]

Table 3.10: Generated Reports

Report Type	Purpose
Visitor Report	Visitor statistics
Approval Report	Approval tracking
Security Report	Entry and exit logs
Audit Report	System activity records
Department Report	Department-wise visitor analysis

3.2.8 Overall Workflow Summary

The overall workflow of the system follows the sequence:

Visitor Registration → Approval Request → Approval Decision → QR Pass Generation → Security Verification → Check-In → Visit Monitoring → Check-Out → Audit Logging → Reporting

This structured workflow ensures secure visitor management, improved operational efficiency, accurate record keeping, and enhanced organizational security.

Section 3.1.1 — Technology Stack Overview

#	Layer	Technology	Version	Purpose
1	Frontend Framework	React.js	18.x	Component-based SPA (Single Page Application)
2	Frontend Build Tool	Vite	7.x	Ultra-fast development server and production bundler
3	Frontend Language	TypeScript	5.x	Type-safe JavaScript for frontend logic
4	UI Styling	Tailwind CSS	4.x	Utility-first responsive styling framework
5	Icon Library	Lucide React	Latest	Consistent, scalable UI icon set
6	QR Code Generator	react-qr-code	Latest	Generates inline SVG QR codes for visitor passes
7	Camera Integration	react-webcam	Latest	Live webcam feed for visitor identity capture
8	Routing	React Router DOM	v6	Client-side SPA navigation with protected routes
9	HTTP Client	Axios	Latest	JWT-authenticated REST API requests
10	Animations	Framer Motion	Latest	Smooth page transitions and micro-animations
11	Backend Framework	FastAPI	0.111.x	High-performance Python REST API with async support
12	Backend Language	Python	3.10+	Core application logic and data processing
13	ORM	SQLAlchemy	2.x	Database model definitions and query abstraction
14	Database	PostgreSQL	15	Primary relational data store
15	Cache / Session	Redis	7	Token caching and real-time session support
16	Authentication	JWT (JSON Web Token)	RFC 7519	Stateless, role-encoded access tokens
17	Password Hashing	Passlib / bcrypt	Latest	Secure one-way password hashing

[FIGURE 3.12 – TECHNICAL STACK DIAGRAM]

3.3 Results and Discussion

The Enterprise Visitor Verification & Access Control Platform was successfully developed and implemented as a centralized web-based solution for managing visitor registration, approval workflows, access control, and reporting operations at India Glycols Limited (IGL). The platform effectively digitizes the traditional visitor management process and provides enhanced security, operational efficiency, transparency, and real-time monitoring capabilities.

The developed system supports multiple user roles, including System Administrators, Plant Administrators, Department Heads, Employees, and Security Officers. The implementation of Role-Based Access Control (RBAC) ensures that each user can access only the functionalities relevant to their responsibilities. This approach improves data security, prevents unauthorized access, and enhances overall system governance.

The Visitor Registration Module enables visitors to register through a self-service kiosk interface by entering their personal information, visit purpose, host details, and other required information. Visitor records are validated and securely stored in the database. The platform also supports visitor photograph capture and document management, eliminating the need for manual paper-based records.

The Approval Workflow Module efficiently routes visitor requests to the appropriate employee or department head for review. Authorized personnel can approve or reject requests directly through the dashboard. This automated workflow significantly reduces approval delays and improves coordination between visitors and host employees.

The QR Code-Based Visitor Pass Generation Module automatically generates secure digital visitor passes after approval. Each pass contains a unique QR code that serves as a digital identifier for verification purposes. The generated passes are print-ready and can be quickly validated by security personnel during entry and exit procedures.

The Security Verification Module enables security officers to verify visitor passes through QR code scanning. This process ensures accurate visitor identification and prevents unauthorized access to organizational facilities. The module also records visitor check-in and check-out activities, providing real-time visibility into visitor movements within the premises.

The Reporting and Analytics Module generates operational reports such as visitor statistics, approval histories, security activity logs, audit records, and department-wise visitor

reports. These reports assist management in monitoring visitor activity, improving security practices, and maintaining compliance with organizational policies.

Table 3.11: Functional Results Achieved

Module	Result Achieved
Visitor Registration	Successful digital visitor onboarding
Approval Workflow	Automated approval processing
QR Pass Generation	Secure digital pass creation
Security Verification	QR-based access validation
Check-In / Check-Out	Accurate visitor tracking
Reporting Module	Real-time reporting and analytics
RBAC System	Secure role-based access control
Audit Logging	Complete activity tracking

The implementation of Docker containerization significantly simplified deployment and environment management. The frontend, backend, database, Redis cache, and supporting services were successfully deployed using Docker containers, ensuring consistency across development, testing, and production environments. Containerized deployment reduced configuration issues and improved maintainability of the overall system.

Table 3.12: Performance Observations

Parameter	Observation
User Authentication	Secure and responsive
Visitor Registration	Fast and user-friendly
Approval Processing	Automated workflow
QR Verification	Accurate and reliable

Database Operations	Efficient data retrieval
Report Generation	Real-time reporting
System Security	Enhanced through JWT and RBAC
Deployment	Simplified through Docker

The developed platform provides several organizational benefits, including reduced paperwork, improved visitor tracking, faster approval processing, centralized record management, enhanced security controls, and better audit compliance. The platform also supports scalability and can be extended across multiple plants and facilities without significant architectural changes.

However, certain limitations remain. The platform currently relies on network connectivity for real-time operations, and some processes still depend on user input accuracy. Future improvements may include mobile application support, biometric verification, visitor pre-registration, automated email and SMS notifications, and integration with physical access control devices.

Table 3.13: Advantages and Future Enhancements

Current Advantages	Future Enhancements
Digital visitor management	Mobile application support
QR-based verification	Biometric integration
Centralized monitoring	Smart access control devices
Automated approvals	SMS and Email notifications
Secure access control	AI-powered analytics
Comprehensive reporting	Multi-site cloud deployment

The system was tested across various visitor management scenarios, including visitor registration, approval of workflows, pass generation, security verification, check-in/check-out operations, and report generation. The results demonstrated stable performance, accurate data processing, and effective workflow automation throughout the visitor lifecycle.

In conclusion, the Enterprise Visitor Verification & Access Control Platform successfully achieved its objectives by providing a secure, scalable, and efficient visitor management solution for India Glycols Limited (IGL). The platform enhances organizational security, streamlines visitor processing, improves operational efficiency, and establishes a strong foundation for future digital transformation initiatives within the organization.

4. Project Outcome and Applicability

4.1 Project Outcome

The Enterprise Visitor Verification & Access Control Platform was successfully designed, developed, and implemented as a comprehensive digital solution for managing visitor entry, approval workflows, access verification, and security monitoring at India Glycols Limited (IGL). The project replaces traditional paper-based visitor management processes with a secure, centralized, and automated platform that improves operational efficiency and organizational security.

One of the primary outcomes of the project is the successful digitization of the visitor registration process. Visitors can register through a self-service kiosk interface by providing personal details, visit purpose, host information, and supporting documents. This eliminates manual entry procedures and reduces administrative workload.

Another significant outcome is the implementation of an automated approval workflow. Visitor requests are routed to the appropriate employee, department head, or authorized personnel for review and approval. This ensures faster processing, improved transparency, and better coordination between visitors and hosts.

The platform also generates secure QR Code-based visitor passes. Each approved visitor receives a unique digital pass containing identification information and a QR code for verification. This enables quick authentication at security checkpoints and minimizes the possibility of unauthorized access.

The system provides real-time visitor tracking through check-in and check-out functionalities. Security personnel can verify visitor identities, monitor entry and exit activities, and maintain accurate visitor logs. This improves facility security and ensures complete visibility of visitor movements within the organization.

A comprehensive reporting and analytics module has also been developed. The platform generates visitor reports, approval statistics, audit logs, department-wise visitor summaries, and security activity reports. These reports support management in monitoring operations, improving security practices, and ensuring compliance with organizational policies.

The project successfully incorporates modern technologies such as React, FastAPI, PostgreSQL, Redis, Docker, JWT Authentication, QR Code Generation, and Role-Based Access Control (RBAC), resulting in a scalable and enterprise-ready solution.

Table 4.1: Major Project Outcomes

Outcome	Description
Digital Visitor Registration	Automated visitor onboarding process
Approval Workflow Automation	Faster and transparent approvals
QR Code Visitor Passes	Secure visitor identification
Real-Time Visitor Tracking	Accurate entry and exit monitoring
Centralized Database	Secure storage of visitor information
Role-Based Access Control	Controlled access for different users
Reporting & Analytics	Operational insights and audit reports
Docker-Based Deployment	Simplified deployment and maintenance

The implementation of the system has resulted in improved security, reduced paperwork, enhanced operational efficiency, and better management of visitor records. The platform serves as a reliable and scalable solution for enterprise-level visitor management requirements.

4.2 Applicability of the System

The Enterprise Visitor Verification & Access Control Platform has broad applicability across industrial facilities, corporate offices, educational institutions, healthcare organizations, government departments, and other environments where visitor management and access control are critical.

A. Industrial Facilities and Manufacturing Plants

Industrial organizations such as India Glycols Limited can use the system to manage visitors, contractors, vendors, and temporary workers. The platform ensures secure access to sensitive operational areas while maintaining complete visitor records for compliance and security purposes.

B. Corporate Offices

Corporate organizations can utilize the system to streamline visitor registration, meeting approvals, employee hosting, and reception management. The platform improves visitor experience while maintaining security standards.

C. Educational Institutions

Universities, colleges, and schools can use the system to manage visitors, parents, guest lecturers, and event participants. The platform provides controlled access and accurate visitor monitoring.

D. Hospitals and Healthcare Facilities

Healthcare institutions can implement the system to manage patient attendants, visitors, contractors, and vendors. Real-time visitor tracking improves safety and ensures compliance with healthcare regulations.

E. Government and Public Sector Organizations

Government offices can use the platform to regulate visitor entry, maintain audit records, and improve security procedures through automated verification and approval mechanisms.

F. Contractor and Vendor Management

The system can be extended to manage contractor access, vendor visits, temporary workforce permissions, and service personnel activities. This ensures proper authorization and monitoring of external personnel.

G. Security and Compliance Management

Organizations can use the platform to maintain complete audit trails, visitor history, approval records, and security logs. These records support compliance audits and internal security reviews.

H. Multi-Location Enterprise Deployment

The platform supports scalability and can be deployed across multiple plants, offices, or facilities. Centralized administration allows organizations to manage visitor activities across different locations through a unified system.

Table 4.2: System Applicability

Domain	Application
Manufacturing Plants	Visitor and contractor management
Corporate Offices	Visitor registration and approvals
Educational Institutions	Campus visitor monitoring
Healthcare Facilities	Patient visitor management
Government Organizations	Secure public access management
Vendor Management	External workforce control
Security Operations	Access verification and monitoring
Multi-Site Enterprises	Centralized visitor administration

The modular architecture of the platform enables future enhancements such as biometric authentication, facial recognition, mobile applications, email and SMS notifications, cloud deployment, AI-based visitor analytics, and integration with physical access control systems. These capabilities make the system suitable for both current operational requirements and future digital transformation initiatives.

In conclusion, the Enterprise Visitor Verification & Access Control Platform provides a secure, scalable, and efficient solution for visitor management. Its applicability across multiple domains and its ability to support future enhancements make it a valuable enterprise solution for modern access control and security management.

5. Conclusion

The Enterprise Visitor Verification & Access Control Platform successfully demonstrates the implementation of a modern, secure, and efficient visitor management solution for India Glycols Limited (IGL). The project was developed to replace traditional paper-based visitor management processes with a centralized digital platform capable of handling visitor registration, approval of workflows, access verification, monitoring, and reporting.

The system effectively streamlines the complete visitor lifecycle, starting from visitor registration and approval to QR code generation, security verification, check-in/check-out tracking, and audit reporting. By automating these processes, the platform significantly reduces manual effort, improves operational efficiency, and enhances organizational security.

One of the major strengths of the system is its Role-Based Access Control (RBAC) mechanism, which ensures that users can access only the functionalities relevant to their responsibilities. The implementation of secure authentication, QR code-based visitor verification, centralized record management, and real-time monitoring contributes to a highly reliable and secure access control environment.

The project also demonstrates the effective use of modern technologies such as React, TypeScript, FastAPI, PostgreSQL, Redis, Docker, JWT Authentication, and QR Code Generation. The modular architecture adopted during development ensures scalability, maintainability, and ease of future enhancements. Docker-based deployment further simplifies system administration and guarantees consistency across development and production environments.

The reporting and analytics capabilities of the platform provide valuable insights into visitor activities, approval trends, security operations, and audit records. These features support management in making informed decisions while maintaining compliance with organizational policies and security standards.

The implementation of the platform has resulted in several benefits, including reduced paperwork, faster visitor processing, improved access control, centralized visitor data management, enhanced security monitoring, and better audit compliance. The system is also flexible enough to support deployment across multiple plants, facilities, and organizational locations.

Although the current implementation successfully fulfills its intended objectives, future enhancements can further increase its capabilities. Potential improvements include mobile application support, biometric authentication, facial recognition integration, automated SMS and email notifications, cloud deployment, visitor pre-registration, and integration with physical access control systems.

In conclusion, the Enterprise Visitor Verification & Access Control Platform provides a secure, scalable, and efficient solution for modern visitor management requirements. The project successfully achieves its objectives by improving security, optimizing visitor handling processes, and supporting digital transformation initiatives at India Glycols Limited.

6. References

1. React Documentation. Meta Platforms Inc. Available at: <https://react.dev/>
2. FastAPI Documentation. Sebastián Ramírez. Available at: <https://fastapi.tiangolo.com/>
3. PostgreSQL Global Development Group. PostgreSQL Documentation. Available at: <https://www.postgresql.org/docs/>
4. Docker Inc. Docker Documentation. Available at: <https://docs.docker.com/>
5. Redis Documentation. Redis Ltd. Available at: <https://redis.io/docs/>
6. TypeScript Documentation. Microsoft Corporation. Available at: <https://www.typescriptlang.org/docs/>
7. Tailwind CSS Documentation. Tailwind Labs. Available at: <https://tailwindcss.com/docs>
8. SQLAlchemy Documentation. Available at: <https://docs.sqlalchemy.org/>
9. JSON Web Token (JWT) Documentation. Available at: <https://jwt.io/>
10. QR Code Generator and Standards Documentation. Available at: <https://www.qrcode.com/en/>
11. Vite Documentation. Available at: <https://vitejs.dev/>
12. Nginx Documentation. Available at: <https://nginx.org/en/docs/>
13. Python Software Foundation. Python Documentation. Available at: <https://docs.python.org/3/>
14. Lucide React Documentation. Available at: <https://lucide.dev/>
15. India Glycols Limited (IGL) – Visitor Management Requirements and Internal Process Documentation.
16. Enterprise Security Best Practices for Access Control Systems. National Institute of Standards and Technology (NIST). Available at: <https://www.nist.gov/>
17. OWASP Foundation. Web Application Security Guidelines. Available at: <https://owasp.org/>
18. PostgreSQL Role-Based Access Control (RBAC) Documentation. Available at: <https://www.postgresql.org/docs/>

7. Contribution

Milind Verma (23BCE11695)

As the sole developer of the Enterprise Visitor Verification & Access Control Platform, I was responsible for the complete design, development, testing, deployment, and documentation of the project. The project was developed during my internship at India Glycols Limited (IGL) with the objective of digitizing and streamlining the visitor management process. I performed a requirement gathering and analyzed the existing visitor entry process to identify challenges and opportunities for automation. Based on the collected requirements, I designed the overall system architecture, database structure, workflow diagrams, and access control mechanisms. I developed the complete frontend application using React, TypeScript, Vite, and Tailwind CSS. This included designing and implementing the Visitor Registration Kiosk, Employee Approval Dashboard, Security Dashboard, Administrative Panels, Reporting Modules, and QR Code-based Visitor Pass System.



I was also responsible for backend development using Fast API. I developed REST APIs for visitor registration, approval of workflows, authentication, authorization, reporting, and visitor tracking. JWT-based authentication and Role-Based Access Control (RBAC) were implemented to ensure system security.

I designed and managed the PostgreSQL database, including tables for users, visitors, approvals, security logs, audit records, and administrative data. I also optimized database operations to ensure efficient data retrieval and storage.

To enhance security and operational efficiency, I implemented QR Code generation and verification features that enable quick visitor validation during entry and exit procedures. I also developed check-in and check-out functionalities for real-time visitor monitoring.

Additionally, I containerized the complete application using Docker and configured the deployment environment for frontend, backend, database, and supporting services. I

performed testing, debugging, and performance optimization to ensure the stability and reliability of the system.

Apart from development activities, I prepared complete project documentation, including system architecture diagrams, database design, workflow diagrams, technical specifications, testing reports, and project reports.

Through this project, I gained practical experience in full-stack web development, enterprise software design, database management, API development, application security, Docker containerization, and real-world software deployment. The successful completion of this project demonstrates my ability to independently develop and deploy a complete enterprise-level software solution.